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Beyond Textual Alchemy: Explainable Diagnostics for LLM Jailbreaks

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Abstract—This paper argues that adversarial vulnerability research in Large Language Models (LLMs) must abandon heuristic alchemy focused solely on attack success rate in favor of mechanistically explainable attacks. Contemporary methodologies, from gradient-based token searches to conversational inductions and automated rewriting, bypass safety guardrails with remarkable efficacy yet remain empirical trials that fail to illuminate why alignment fails. We propose that ***a scientifically rigorous adversarial attack must satisfy three core dimensions: mechanistic attribution (localizing failure circuits), causal mapping (interventionally linking perturbations to latent-space shifts), and constructive actionability.*** *By reframing attacks as diagnostic instruments rather than mere bypassing tools, we can provide precise algorithmic targets for fundamental model interventions. This framework shift enables the community to transition from superficial application-layer filtering to targeted mechanistic editing and robust representation-level realignment, which is essential for engineering verifiably secure intelligent systems.*

Index Terms: *Large language models, jailbreak attacks, mechanistic interpretability, adversarial robustness, explainable diagnostics, AI safety.*

Large Language Models (LLMs) have rapidly evolved from isolated natural language processors into the foundational cognitive engines of pervasive intelligent systems, driving innovations across diverse domains. To ensure these models operate within ethical boundaries and human values, developers rely heavily on safety alignment protocols, such as Reinforcement Learning from Human Feedback (RLHF)[1] and Direct Preference Optimization (DPO).[2] These techniques aim to construct a robust refusal landscape that prevents the generation of harmful or illicit content. Despite rigorous application, alignment remains far from definitive. Rather than fundamentally neutralizing unsafe knowledge, these techniques often merely instill a layer of superficial behavioral mimicry, a safety wrapper that masks but does not erase the model's underlying vulnerabilities.[3] This inherent incompleteness has been systematically exposed by the emergence of jailbreak attacks. By exploiting the residual tension between a model's latent capabilities and its aligned refusal tendencies, these adversarial perturbations successfully peel back the safety wrapper, revealing that current guardrails can be bypassed through targeted manipulation of the input space.

Current vulnerability discovery is dominated by heuristic optimization and empirical trial-and-error. Methodologies utilizing gradient-based token search,[4] heuristic algorithms, and LLM-assisted prompt rewrit-

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ing[5,6] have achieved remarkable success in bypassing alignment filters. Yet, this success is almost exclusively measured by a single, phenomenological oracle: the Attack Success Rate (ASR). By prioritizing ASR over mechanistic understanding, the security community has inadvertently embraced a culture of textual alchemy. Attackers treat the target model as an opaque black box, injecting nonsensical suffix strings or randomly mutating role-play scenarios until the model stochastically produces a harmful output.

This purely empirical approach creates a dangerous illusion of progress. While high ASRs successfully expose that the safety manifold has collapsed, they fundamentally fail to articulate *why*. Without tracking the continuous trajectory of how an input prompt navigates the high-dimensional, non-convex refusal landscape, these attacks offer no characterization of the refusal boundary's structure, and hence no smoothness or margin property on which principled defenses could rely. Consequently, the response to a novel jailbreak is often limited to reactive data patching and superficial fine-tuning. This "whack-a-mole" defense strategy is ultimately unsustainable. It incurs exorbitant computational overhead, exacerbates the alignment tax, and degrades model utility through catastrophic forgetting, over-refusal, and a loss of helpfulness. As the field advances toward multimodal and agentic systems, the attack surface grows rapidly. In these complex environments, heuristic trial-and-error will soon become computationally intractable. To advance the science of AI security from empirical alchemy to rigorous engineering, attacks must become high-fidelity diagnostic instruments rather than mere exploitation tools. This agenda deliberately builds on the interpretability, alignment, and security literatures rather than claiming to supersede them; its contribution is integrative: unifying their tools into a diagnostic protocol, rendering its dimensions measurable, and restructuring the research agenda around repair.

To conceptually formalize this transition, **Figure 1** contrasts the prevailing empirical approach with our proposed diagnostic framework, wherein an *Explainable Adversarial Attack* is strictly defined and evaluated across three core dimensions. The first, *mechanistic attribution*, demands the rigorous localization of the specific attention heads or neural circuits where safety suppression mechanisms fail. The second, *causal mapping*, requires quantifying how semantic or syntactic perturbations dynamically shift the model's internal state off the aligned safety manifold. Finally, *constructive actionability* dictates that these diagnostic insights must translate directly into structural defenses. By revealing the coordinates of a vulnerability, such attacks cease to be destructive anomalies and instead provide topological priors for representation-level realignment, such as targeted gradient regularization or mechanistic editing, that aim to neutralize adversarial pathways at their root.

The Interpretability Gap: Why ASR is Insufficient for Safer LLMs

The contemporary landscape of LLM adversarial research is evaluated by a single metric: the Attack Success Rate (ASR). The literature's overarching objective has been to discover prompts that circumvent safety alignment, a campaign that has yielded a valuable taxonomy of jailbreak methods and proven that alignment mechanisms are fragile, yet offers limited mechanistic insight for developing safer models or patching architectural flaws. The limitations of this ASR-centric framework are not merely academic. By focusing on the outcome rather than the process, the community has prioritized exploitation over diagnosis: benchmarks treat LLMs as monolithic black boxes that record a binary success or failure, masking the high-dimensional dynamics within the model's latent representations. Without understanding which circuits are hijacked during a successful jailbreak, defenders fall back on reactive patching with superficial filters that the next generation of adaptive attacks easily bypasses. None of this argues for discarding ASR. It remains the indispensable screening metric that establishes whether a vulnerability exists and anchors comparability across benchmarks; the claim is that ASR is necessary but insufficient, and must be complemented by diagnostic metrics that characterize why a failure occurs and how it may be repaired.

Furthermore, the lack of mechanistic interpretability in current attack methodologies prevents the development of principled defenses. If an attack does not explain how it circumvents a safety guardrail, it cannot inform targeted interventions, such as weight adjustments or representation-level realignment, aimed at addressing the underlying flaw. This section examines the interpretability gaps of these attack modalities, arguing that adversarial research must transition from proving the existence of flaws to mapping their internal causal mechanisms.

Discrete Optimization: Exposing Fragility Without Mechanistic Explanation

The most prominent class of white-box attacks relies on token-level gradients to search for adversarial suffixes. Rooted in early prompt tuning, this framework

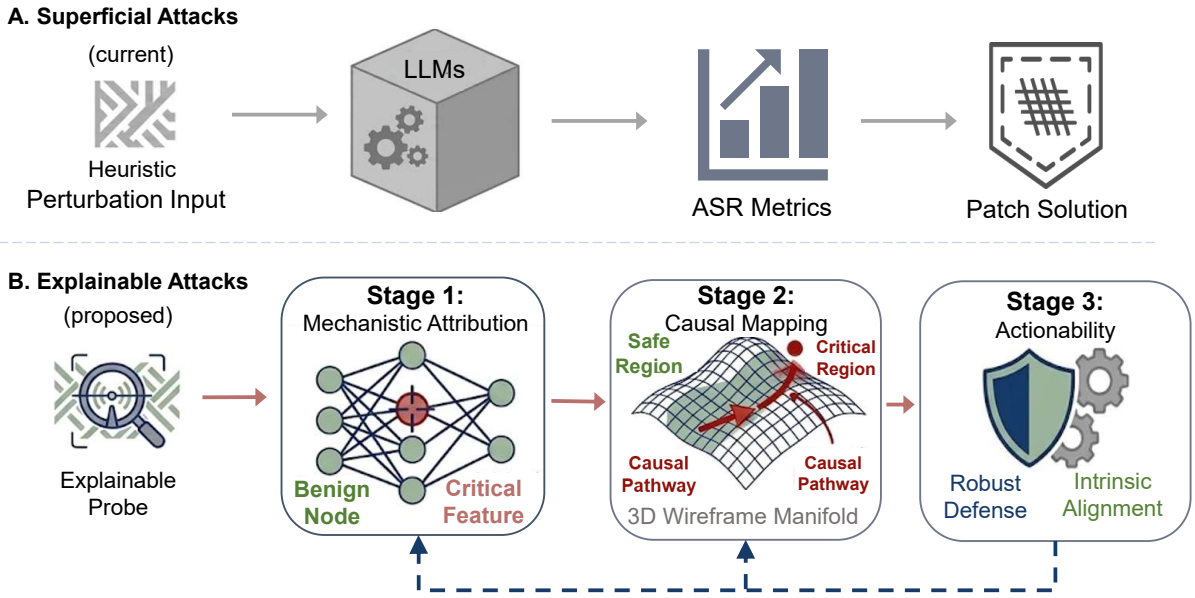


FIGURE 1. Shifting from phenomenological exploitation toward explainable adversarial attacks (a schematic illustration of the contrast between one line of existing attack research and our proposal, not a survey of the whole field). (A) A line of existing work treats the LLM as an opaque black box, optimizes heuristically for ASR, and responds with superficial whack-a-mole patching. (B) We propose a diagnostic framework defined across three dimensions: **Dim 1** strictly attributes vulnerabilities to specific neural circuits; **Dim 2** maps the causal trajectory of how perturbations shift representations off the safety manifold; **Dim 3** translates these coordinates into structural interventions aimed at neutralizing adversarial pathways at their root, with the diagnostic loop iterating as interventions are re-evaluated.

was propelled to the forefront by the seminal Greedy Coordinate Gradient (GCG) attack[4] and its accelerated variants. Undeniably, these methodologies have made a framework-shifting contribution to LLM security. By framing jailbreaking as a targeted maximum likelihood estimation problem, they provided mathematically rigorous proofs that aligned LLMs harbor contiguous adversarial vulnerabilities within their discrete token space. This foundational work effectively shattered the illusion of robust safety alignment, convincingly demonstrating that alignment often acts merely as a superficial behavioral wrapper rather than a fundamental restructuring of the model's internal knowledge representation.

However, while these optimization-based attacks excel at exposing the existence of vulnerabilities, their efficacy is entirely decoupled from coherent semantic logic. Because they rely on greedy coordinate descent to navigate a highly non-convex refusal landscape, the resulting adversarial sequences are almost exclusively characterized by extreme semantic fragmentation and high perplexity.[4] These gibberish suffixes act merely as brittle mathematical keys that happen to unlock a specific door. From a mechanistic standpoint, a GCG suffix constitutes an interpretability black hole. It offers zero insight into the causal mechanics of the breach. It remains entirely opaque whether these arbitrary combinations of meaningless characters succeed by dynamically suppressing specific refusal attention heads, hijacking the in-context learning circuitry, or merely exploiting out-of-distribution (OOD) logit shifts. As explicitly noted by the *JailbreakLens* study,[7] the understanding of underlying mechanisms remains limited, reducing these discrete attacks to empirical observations rather than diagnostic tools.

Semantic and Contextual Black-Box Attacks: The Explanatory Void

To overcome the readability and perplexity limitations inherent in discrete gradient-based optimization, the research community has successfully pioneered semantic and contextual manipulation strategies.[8] This

shift has yielded two highly impactful frameworks. First, prompt-to-prompt adversarial frameworks (e.g., PAIR,[5] TAP[6]) employ auxiliary LLMs to dynamically rewrite source queries into sophisticated linguistic deceptions, mapping a vast landscape of Out-of-Distribution (OOD) semantic vulnerabilities. Second, attacks exploiting the In-Context Learning (ICL) capabilities of LLMs, most notably Many-Shot Jailbreaking and its context-curated variants, have demonstrated that systematically saturating the context window with fabricated input-output pairs can effectively dismantle alignment safeguards. These methods have made substantial contributions to the field. By simulating realistic, human-like adversaries and weaponizing the model's core adaptive strengths, they have convincingly proven that current safety alignment is remarkably fragile under coherent, long-context stress tests.

However, while these heuristic and contextual strategies excel at exposing the breadth of the vulnerability surface and consistently achieving ASR, they collectively fall into a profound explanatory void. Employing one black-box model to probe another, or artificially inflating context windows with malicious demonstrations, inherently obscures the causal chain of the attack. When a roleplay trick or many-shot injection bypasses safety filters, why it succeeded remains mechanistically opaque. This reliance on end-to-end empirical success metrics severely limits progress in representation engineering. Current analyses cannot tell whether the breach stems from the dynamic suppression of internal refusal attention heads, the exploitation of flaws in in-context reasoning circuitry, or the artificial inflation of the probability mass of malicious tokens via simple induction head mechanisms. By prioritizing the discovery of novel semantic bypasses over the diagnosis of internal representational shifts, these approaches yield a what without a why.

This dual explanatory void cascades into a single, severe defensive dilemma. Because the community cannot trace how adversarial inputs, whether gibberish suffixes or semantic manipulations, hijack the model's internal machinery, defenders are relegated to reactive, application-layer patching: perplexity filters, input randomization,[9] hard-coded prompt blacklists, or aggressive context truncation. Such heuristics overfit the decision boundary to specific, localized attack clusters, and their apparent success often reflects the weakness of the optimizers they counter rather than any fundamental repair of the model's safety representations. Defenses thus act as band-aids, and the cycle grows more urgent as successful suffixes transfer across models and adapt around mitigations. Breaking this cycle requires the diagnostic transition developed in the remainder of this article: from cataloging what breaks an LLM to mapping how an optimization trajectory collapses its internal safety manifold.

Transitioning from Exploitation to Diagnosis

The pervasive reliance on ASR has mobilized the community to acknowledge the fragility of LLM alignment,[10] and heuristic attacks served as necessary wake-up calls. But ASR is a purely phenomenological metric: it records a behavioral failure while remaining silent on its mechanistic cause. As LLMs move into mission-critical deployment, this approach becomes scientifically insufficient, and adversarial vulnerability research must shift its objective function from maximizing exploitation to maximizing diagnosis.

This transition demands a new framework: **Explainable Adversarial Attacks**. To be scientifically rigorous, an attack must transcend binary success/failure outcomes and output a verifiable causal trace of the model's internal state transitions; only by diagnosing the neural circuitry behind a breach can we inform the targeted interventions required for representation engineering. The following section formalizes this framework.

Defining the Explainable Adversarial Attack

Given an insufficient phenomenological framework, a rigorous foundation is needed: an Explainable Adversarial Attack must satisfy mechanistic attribution, causal mapping, and constructive actionability.

Positioning within current research. None of these dimensions requires inventing interpretability from scratch; recent years have produced a substantial toolkit that a diagnostic program can build upon. Sparse autoencoders and feature decomposition now extract interpretable features at frontier scale,[11] giving attribution a vocabulary in which failures can be stated. Activation steering[12,13] shows that behavior can be redirected at inference time by editing activation directions, precisely the intervention class constructive actionability demands. Circuit discovery[14] and causal mediation analysis[15] supply the verification machinery that separates attribution from correlation, and refusal behavior is a uniquely tractable target for this toolkit, as it appears mediated by a small set of direction-like components.[16] The framework's contribution is thus integrative rather than methodological: it binds these components into a single diagnostic protocol with explicit evaluation criteria and insists their outputs be validated causally rather than accepted as correlations.

The tools' limitations, notably scalability, polysemanticity, and restricted access to proprietary models, are substantial; we return to them below, and they define the feasibility frontier against which the agenda of the following section must be read.

Mechanistic Attribution: Localizing the Neural Circuits of Failure

The first dimension of our proposed framework requires that an adversarial attack transcend the binary metric of success or failure to provide a localized causal trace of the internal breakdown. In current security literature, safety alignment is often treated as a monolithic barrier. However, recent advances in mechanistic interpretability reveal that alignment is not a singular safety gate but rather a capability sparsely distributed across specific neural circuits, such as high-impact attention heads and localized latent subspaces.

Because safety mechanisms are fundamentally decentralized, an explainable adversarial attack must pinpoint the neural pathways where these mechanisms are subverted. To achieve this, the attack protocol must intrinsically couple the generation of adversarial inputs with continuous activation tracking and causal interventions. By tracing the forward pass, this methodology allows researchers to classify candidate failure modes. For instance, it provides the diagnostic resolution to determine whether an attack succeeds by dynamically suppressing refusal attention heads that govern constraint satisfaction or by exploiting residual stream bottlenecks to circumvent safety-relevant features entirely.

By enforcing this level of mechanistic attribution, we fundamentally shift the unit of analytical interest from the superficial adversarial input to the underlying compromised neural circuit. When a successful jailbreak can be causally linked to the anomalous activation of an identifiable subset of attention heads, such as those responsible for intent parsing or persona shifting, it yields a high-fidelity diagnostic map of the model's vulnerability. This granular localization fulfills the first theoretical requirement of our framework: transforming an abstract behavioral breach into candidate structural coordinates.

Causal Mapping: Establishing the Input-to-Representation Link

The second dimension of our framework formalizes the discrete-to-continuous bridge. An explainable adversarial attack must establish a verifiable causal chain linking specific, discrete input-level perturbations to quantifiable, continuous shifts within the model's high-dimensional latent space.

To satisfy this criterion, the attack methodology must model the LLM's *refusal landscape*: a topologically bounded safety manifold within the latent space, from which a successful jailbreak is a directed transition into an unsafe activation region. Causal mapping traces this trajectory and identifies the critical boundaries where the model's representation shifts from constraint satisfaction to malicious execution.

This mapping must be anchored by quantitative divergence metrics. Rather than relying on end-to-end discrete outcomes, the attack tracks layer-wise trajectories, such as maximum mean discrepancy or cosine distance, against a benign baseline, quantifying the degree of representation misdirection and determining whether the bypass relies on abrupt numerical anomalies or a smooth, gradual collapse across the safety boundary. This geometric map connects attribution and repair: it explains how the input perturbation manipulates the localized circuits, supplying candidate coordinates for constructive repair.

Constructive Actionability: From Diagnosis to Principled Repair

The ultimate metric of an explainable adversarial attack is its constructive actionability: the degree to which its diagnostic output can directly guide foundational model repair. Whereas traditional discovery ends at documenting an exploit, an explainable attack uses the mapped coordinates to enable principled, mechanistic interventions.

By satisfying the criteria of mechanistic attribution and causal mapping, the attack yields a high-fidelity blueprint of the model's failure logic. This continuous diagnostic data empowers developers to transcend broad, data-driven fine-tuning and implement targeted, mechanism-grounded defensive strategies:

- **Activation Steering**: Using the mapped trajectories to design real-time, inference-time interventions: applying counter-vectors steers latent activations away from malicious sub-manifolds and back toward the refusal landscape.
- **Targeted Model Editing**: Leveraging attribution to update only the weights of compromised circuits, aiming to excise vulnerabilities at their root while circumventing the catastrophic forgetting of broad retraining.
- **Representation-level Regularization**: Encoding discovered failure paths as geometric regularization constraints during alignment or unlearning, enforcing smoothness across refusal boundaries and preventing the abrupt representational shifts that adversarial inputs exploit.

In this framework, the explainable attack is redefined: no longer a destructive exploit but a verifiably robust diagnostic instrument, the foundational prerequisite for engineering resilient intelligent systems.

From Attack to Diagnostic Report: A Five-Step Protocol

To make the three dimensions implementable, we outline a five-step protocol that converts an arbitrary jailbreak attempt into a structured diagnostic report. It assumes white-box access to an open-weight model; adaptations for proprietary systems are discussed later. **Activation acquisition.** The analyst hooks the residual stream and attention heads at every layer while running the benign prompt and its adversarial variant, producing aligned layer-wise activation tensors. **Attribution screening.** Components whose states shift systematically under the attack are identified through attention-head contribution analysis, sparse-autoencoder feature activation,[11] or causal mediation,[15] yielding a candidate component set A. **Causal verification.** Activation patching or interchange interventions[14] test whether manipulating the candidates restores refusal on the adversarial input; only components that pass are retained as the verified set A^*, and the pass rate over the attack corpus is the Counterfactual Verification Rate defined in Table 1. **Trajectory mapping.** Layer-wise divergence (cosine distance or maximum mean discrepancy) between adversarial and benign trajectories locates the peak layers where the state crosses the aligned boundary, producing a Trajectory Divergence profile. **Intervention synthesis.** These verified coordinates are translated into candidate defenses, namely steering vectors from the localized refusal direction, editing targets from A^*, or boundary-derived regularization constraints, and assembled into a diagnostic report together with the computed metrics and a prioritized intervention menu.

Each step is executable with existing open-source tooling: activation hooking is standard in interpretability libraries, activation-patching tooling covers the causal verification step, and public sparse autoencoders now exist for open-weight frontier models. The outputs are precisely the quantities the benchmark metrics of the next section measure, closing the loop between the framework's principles and its evaluation. The protocol's computational cost, and its graceful degradation for closed models, are discussed below.

Worked Example: Re-Analyzing GCG under the Diagnostic Framework

To illustrate how the framework changes research practice, we walk through a concrete re-analysis of the Greedy Coordinate Gradient (GCG) attack,[4] the canonical white-box suffix attack, following the operational protocol just described. Throughout, we deliberately distinguish evidence already established in the literature from protocol steps that remain open research targets, so that the reader can assess exactly what the framework promises today versus what it demands of future work.

Stage 1: Mechanistic attribution of a GCG suffix. Following the protocol, the analyst records layer-wise activations for a benign prompt and its adversarial counterpart, screens for components whose states shift systematically under the suffix, and refines candidates through sparse feature attribution[11] and attention-head contribution analysis, in the spirit of recent studies that interpret jailbreaks through representation and circuit lenses.[7] Two findings make GCG a tractable testbed. First, refusal behavior in aligned LLMs is mediated by a comparatively small set of direction-like components,[13,16] which bounds the search space of candidate circuits. Second, despite their high perplexity, GCG suffixes may be processed by the model as coherent content rather than noise; whether they exploit the same semantic machinery as natural-language jailbreaks is a hypothesis that the counterfactual test below can settle. Crucially, correlation is not treated as attribution: a candidate circuit is deemed mechanistically sufficient only after counterfactual verification, that is, when patching or ablating the identified components restores refusal on the adversarial input, a causal test inherited from activation patching and causal mediation practice.[14,15] This operational criterion is precisely what separates causal attribution from mere activation correlation.

Stage 2: Causal mapping of the refusal trajectory. With candidate circuits localized, the protocol tracks how the adversarial input displaces the model's latent state relative to a benign baseline. Layer-wise divergence trajectories (cosine distance or maximum mean discrepancy) locate the layers at which the adversarial trajectory crosses the boundary of the aligned region, yielding a quantitative trace of the attack's progression. For GCG this mapping adjudicates between competing failure hypotheses: a suffix may succeed through a smooth, progressive collapse of the refusal state or through abrupt out-of-distribution logit shifts, and the two hypotheses imply distinct divergence profiles. Recent representation-space analyses of jailbreaks suggest that such boundary transitions

are in principle measurable; formalizing the safety manifold itself, however, remains an open problem, and we regard this stage as the least mature dimension of the framework.

Stage 3: From diagnostic coordinates to structural defense. The value of the preceding stages is that each output translates into a candidate intervention, whereas an ASR report terminates at the failure itself. A verified refusal direction yields an activation-steering defense: counter-vectors applied at the implicated layers during inference restore refusal behavior, an intervention family with growing precedent.[12,13,16] A causally verified circuit yields candidate editing targets or representation-level regularization constraints during alignment. Mechanism-grounded defenses directly address the central weakness of surface defenses such as perplexity filtering and input randomization: because those defenses never engage the underlying failure circuit, adaptive attacks can circumvent them, and the patching itself can degrade model utility. A diagnostic defense instead targets the circuit the attack actually exploits and should therefore transfer across attack variants that share the same failure mechanism, a testable hypothesis and one of the framework's central empirical commitments.

A secondary illustration: many-shot jailbreaking. The framework extends naturally to contextual attacks. Many-shot jailbreaking[17] erodes refusal by saturating the context with fabricated demonstrations, an effect attainable with few in-context examples. Attribution would localize the in-context learning circuitry, induction-style components of the kind identified in canonical circuit analyses, whose activity shifts as demonstrations accumulate; causal mapping would trace the per-demonstration drift of the latent state toward the unsafe region; and actionability would translate this trace into design guidance, such as monitoring the implicated circuitry as an inference-time safety signal or bounding the influence of low-priority context on refusal-critical components.

Taken together, the example demonstrates the framework's practical content: a known attack is decomposed into a set of causally verified failure coordinates, a quantitative trajectory through the refusal landscape, and mechanism-grounded interventions that surface defenses cannot offer. We stress that it is a protocol illustration rather than a completed study: the attribution and steering components rest on published evidence,[7,16] whereas manifold-boundary computation and end-to-end validation of diagnostic defenses remain agenda items. Enforcing this distinction is exactly what the diagnostic metrics proposed below are designed to measure.

A New Research Agenda: From Exploitation to Intrinsic Security

Operationalizing this transition requires restructuring the AI safety research agenda: redefining adversarial attacks as explainable diagnostic protocols that move beyond text-based ASR toward the emerging complexities of next-generation intelligent systems. We propose a three-pronged agenda: (1) Multimodal Diagnostics, tracing causal failures across cross-attention manifolds; (2) System-level Auditing, probing the routing logic of integrated pervasive architectures; and (3) Redefining Benchmarks, establishing computable metrics that mandate mechanistic depth over empirical breadth.

Beyond Text: Mechanistic Diagnostics in Multimodal Manifolds

Visual jailbreaking injects adversarial perturbations into images to trigger malicious text, yet remains more opaque than unimodal attacks:[18] the black box now includes the cross-modal alignment mechanism itself. In this new agenda, explainable attacks must act as diagnostic probes for cross-modal alignment, deconstructing how visual features hijack the textual residual stream through cross-attention layers. Rather than merely recording that a perturbed image causes a failure, a diagnostic attack maps the causal trajectory by which the visual encoder's output shifts the textual latent state off its safety manifold, and determines whether the vulnerability lies in modality-specific encoders or in fusion layers that fail to preserve the semantic consistency of cross-modal representations. Such mechanistic maps move the field beyond image-level filters toward alignment-robust architectures that resist cross-modal adversarial steering by design.

Auditing the Logic: Security in Integrated Pervasive Systems

In pervasive computing environments, LLMs are embedded in multi-component architectures such as Retrieval-Augmented Generation (RAG) and autonomous agents, shifting the threat model from prompt evasion to subversion of operational logic: in an intent-based routing hierarchy, for instance, an attack might target the routing mechanism itself to bypass high-level security policies. A diagnostic agenda must therefore treat integrated systems as holistic reasoning chains, with explainable attacks serving as intent auditors that reveal when routing logic fails to distinguish benign intent from adversarial exploitation. Rather than merely proving that a RAG system can be poisoned, di-

agnostic attacks should trace how adversarial noise in the retrieval pipeline pollutes the internal decision tree and pinpoint the logical nodes or attention mechanisms that fail to filter malicious context. This moves the field from post-hoc output monitoring toward systems whose security is inherent to their integration and intent-verification logic.

Redefining Benchmarks: Metrics for Diagnostic Fidelity and Readiness

To catalyze this framework shift, the research community must move away from the static ASR oracle and establish a new class of evaluative metrics that prioritize mechanistic depth over empirical breadth.[19] Current benchmarks, while instrumental in identifying vulnerabilities, primarily reward the discovery of successful exploits without demanding an explanation of their root cause. This creates a strategic incentive for benchmark hacking, where researchers optimize for success against specific, frozen prompt sets rather than uncovering fundamental flaws in the model's internal logic. We call for the development of diagnostic benchmarks that evaluate adversarial research based on two rigorous new dimensions: diagnostic fidelity and intervention readiness.

Diagnostic Fidelity should quantify the precision, reliability, and interpretability with which an attack localizes a failure circuit or maps a representational shift. In this framework, an attack that provides a verifiable causal trace, linking a specific semantic perturbation to a predictable internal state transition, should be valued more highly than one that achieves success through stochastic trial-and-error. We propose the introduction of localization specificity as a key sub-metric, measuring the overlap between the attack's identified malicious neurons and the model's known safety-critical components. High diagnostic fidelity implies that the attack acts as a high-resolution microscope, revealing not just that a jailbreak occurred, but exactly how the adversarial noise navigated the high-dimensional latent manifold to bypass safety constraints. This metric would effectively penalize gibberish attacks that exploit numerical noise while rewarding those that expose systematic architectural weaknesses.

Intervention Readiness should assess the direct utility of the attack's output for informing diverse downstream security protocols. A high-quality adversarial discovery should not merely reveal a hole in the model's defense; it should provide actionable coordinates precise enough to guide verifiable patching. This metric evaluates the actionability of the diagnostic data across the principled repair modalities defined in our framework:

- **Mechanistic Realignment Readiness**: The degree to which the identified failure trajectory can be used to define new optimization targets for model editing or fine-tuning.
- **Steering Compatibility**: Whether the attack localizes specific activation patterns that can be directly monitored and suppressed by real-time, inference-time steering mechanisms.
- **Boundary Actionability**: The capacity of the attack to define the precise topological boundaries of the safety manifold, allowing for the construction of more robust latent-space guardrails.

TABLE 1. Operational metrics for the proposed diagnostic evaluation.

Metric	Operational definition
Localization Specificity (*LS*)	$LS = \frac{\lvert A \cap S \rvert}{\lvert A \cup S \rvert}$, where A is the attributed component set and S an independently derived set of safety-critical components.
Counterfactual Verification Rate (*CVR*)	$CVR = \frac{\lvert \{x \in X \mid \rho(A,x)=1\} \rvert}{\lvert X \rvert}$, where X is the attack corpus and $\rho(A, x) = 1$ if patching or steering the attributed set A restores refusal on input x; the causal criterion separating attribution from correlation.
Trajectory Divergence (*TD*)	$TD = \max_l d(z_l^{\text{adv}}, z_l^{\text{ben}})$, reported with the peak layer l^*, where d is a layer-wise divergence (cosine distance or maximum mean discrepancy) between adversarial and benign latent trajectories; discriminates smooth manifold crossing from abrupt out-of-distribution shifts.
Intervention Readiness (*IR*)	$IR = w_1 R + w_2 S + w_3 B$, with $w_1 + w_2 + w_3 = 1$, where R, S, and $B \in [0, 1]$ score mechanistic realignment readiness, steering compatibility, and boundary actionability, respectively.

To make these notions operational, Table 1 summarizes candidate metrics that can be computed on existing architectures with current interpretability tooling;[20] the worked example in the preceding section illustrates how several of them arise in practice. Diagnostic Fidelity aggregates the localization, counterfactual, and trajectory components, while Intervention Readiness is the composite *IR*.

Each Intervention Readiness sub-dimension is measured on the same attack corpus used for CVR and maps to one of the repair modalities listed above. Steering compatibility (S) is the refusal-restoration rate obtained by inference-time steering alone, adding the counter-vector without changing any weights; it tests the steering modality specifically, whereas CVR validates attribution through any intervention. Mechanistic

realignment readiness (R) is the restoration rate when repair is restricted to the verified components A^*, multiplied by the fraction of benign-task utility retained. Boundary actionability (B) is how cleanly the peak layer separates safe from unsafe states, e.g., the AUROC of a refusal-direction projection at that layer. Diagnostic Fidelity aggregates the localization, counterfactual, and trajectory components after rescaling each to $[0, 1]$, e.g., $DF = (LS + CVR + TD')/3$ with TD' the normalized divergence; the IR weights are chosen by the evaluator, with equal weights as the default. The set S used by LS should be derived independently of A, for example from published refusal-circuit or steering analyses of the target model family.

These metrics embody two design principles. First, they evaluate the diagnostic quality of an attack rather than its outcome: an attack that achieves a high ASR but fails counterfactual verification receives a low diagnostic score, whereas one that exposes a systematic architectural weakness scores highly regardless of its raw success rate. Second, they complement rather than replace existing security evaluation. ASR remains the screening metric that establishes that a vulnerability exists, while diagnostic metrics characterize why it exists and how it can be repaired. We therefore envision a two-stage evaluation pipeline, in which ASR-based screening identifies candidate attacks and the metrics above profile those that merit mechanistic follow-up, preserving the comparability of ASR benchmarks while incentivizing the mechanistic depth our framework requires.

By rewarding actionability and mechanistic depth, diagnostic benchmarks transform the red team from exploiters of bugs into security architects, ensuring that every discovered vulnerability becomes a stepping stone toward verifiably secure systems and moving AI safety from an empirical cat-and-mouse game toward a principled engineering discipline.

Alternative Views and Discussion

We recognize that this transition involves significant practical and philosophical recalibrations; in the spirit of academic pluralism, we engage with the strongest counterarguments and their implications for AI safety.

The Black-Box Constraint: From Obscurity to Auditable Transparency

Concern.

A primary critique is the commercial reality: state-of-the-art models are predominantly proprietary and accessed via restricted APIs, leaving third-party auditors with ASR as the only seemingly feasible metric. Demanding mechanistic attribution, critics argue, is a non-starter for closed-source deployments.

Response.

We contend that ASR alone creates a security-through-obscurity trap: when failures are observed only at the output layer, neither developer nor auditor understands the underlying vulnerability, and a false sense of security can be shattered by a slight distributional shift. Rather than accepting the black-box constraint as immutable, our framework argues for Diagnostic APIs that let authorized auditors query localized influence signals or activation norms without exposing weights. Explainable diagnostics is not a rejection of black-box testing but a demand for the transparency that high-stakes AI auditing requires.

Until Diagnostic APIs become standard, a layered strategy keeps the framework applicable to closed models: complete white-box diagnosis on open-weight models that share the same alignment mechanisms; test whether localized circuits transfer by probing surrogate behaviors and attack-transfer rates; and, for the closed model itself, exploit every observable signal, including input-output pairs, token probabilities, refusal patterns, and any embedding or logit access, to construct behavioral signatures of the same failure modes. Where regulation permits, negotiated diagnostic endpoints can provide direct influence signals. This stratification converts the black-box constraint from a categorical objection into a graded feasibility question, keeping the framework actionable for proprietary systems.

Resource Allocation: Prioritizing Amortized Safety Gains

Concern.

From an engineering perspective, the computational overhead, often called the interpretability tax, is a legitimate concern. Heuristic attacks are efficient precisely because they optimize directly for a successful string without regard for internal mechanics, whereas mechanistic diagnostics require tracking high-dimensional activation trajectories and intervening on specific circuits, demanding substantially more compute. Critics warn that this disparity could throttle vulnerability discovery and leave models exposed to rapidly evolving lightweight threats.

Response.

We acknowledge this overhead but argue that measuring efficiency solely by discovery speed is myopic. Fast, low-information heuristic attacks incur a hidden, compounding cost: they generate isolated data points

that require reactive patching and yield little reusable knowledge, whereas a single mechanistic diagnosis often exposes a systemic failure mode that can then be neutralized across the board. This is the engineering evolution from brute-force stress testing to structural failure analysis: the upfront cost of explainability is an investment in a scalable, proactive, and principled AI security discipline.

Navigating the Complexity of Neural Representations

Concern.

Finally, large-scale neural representations are high-dimensional and inherently polysemantic: in billion-parameter models, individual neurons encode multiple unrelated concepts, severely complicating precise attribution. Critics legitimately fear selection bias: prioritizing explainable vulnerabilities may over-index on simple, easily visualized failure modes while neglecting insidious, emergent threats that arise from the interplay of thousands of latent dimensions. These concerns are compounded by practical realities: circuit discovery remains computationally expensive, attribution studies often lack ground truth for validation, and findings may not transfer across seeds, checkpoints, or model families.

Response.

This risk is real, but retreating to a purely empirical framework offers no defense against such threats. Vulnerabilities that currently evade mechanistic interpretation do not reflect an inherently unknowable property of models; they expose the limits of our diagnostic toolset. By steadfastly demanding attribution, we incentivize higher-resolution analytical frameworks, such as sparse autoencoders explicitly designed to disentangle polysemantic features, that equip us to decode exactly the stochastic interactions critics fear we might overlook. Accordingly, the framework treats attribution as provisional until it passes counterfactual and stability checks; brittleness is itself a diagnostic finding. The pursuit of explainability is thus not a reductive simplification of model complexity but the commitment to rigor required to transform AI safety from heuristic observation into a mature engineering discipline.

Conclusion

The rapid integration of LLMs has outpaced our understanding of their vulnerabilities: AI safety has long been trapped in a cycle of textual alchemy, measuring heuristic exploits solely by ASR and leaving the mechanistic roots of alignment collapse in the dark.

To break this reactive cycle, we propose a definitive shift toward **Explainable Adversarial Attacks**. By redefining attacks as *diagnostic protocols* rather than mere exploits, our framework integrates three core pillars. It demands *mechanistic attribution* to isolate compromised neural circuits, *causal mapping* to trace how these failures propagate, and *constructive actionability* to ensure these insights yield practical defenses. Under this paradigm, every discovered jailbreak transcends a simple alignment bypass, serving instead as candidate coordinates for principled model repair.

The era of black-box empirical testing must end: as AI evolves into complex agentic systems, diagnosing the why behind a vulnerability is the only sustainable path to intrinsic security and to systems that are secure by design rather than aligned by chance.

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